

Exploratory Data Analysis

Aim

To perform Exploratory Data Analysis (EDA) in Python to understand the structure, relationships, and key patterns present in the dataset.

Procedure

1. Import the required libraries such as pandas, numpy, matplotlib.pyplot, and seaborn for data manipulation and visualization.
2. Load the dataset into a pandas DataFrame.
3. Display the first few rows using head() to get an initial overview of the data.
4. Use functions like info() and describe() to examine data types, non-null counts, and statistical summaries.
5. Check for missing values using isnull().sum() and decide whether to fill or drop them based on data context.
6. Identify duplicate records and remove them if necessary to maintain data quality.
7. Analyze the distribution of individual features using histograms or sns.displot() to understand spread and skewness.
8. Use boxplots to detect outliers in numerical features.
9. Explore relationships between variables using scatter plots, pair plots, or correlation heatmaps.
10. Summarize insights from the analysis such as trends, correlations, and anomalies that will guide further model building.

```
import seaborn as sns
import pandas as pd
import numpy as np
tips=sns.load_dataset('tips')
tips.head()
```

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

```
tips.describe()
```

	total_bill	tip	size
count	244.000000	244.000000	244.000000
mean	19.785943	2.998279	2.569672
std	8.902412	1.383638	0.951100
min	3.070000	1.000000	1.000000
25%	13.347500	2.000000	2.000000
50%	17.795000	2.900000	2.000000
75%	24.127500	3.562500	3.000000
max	50.810000	10.000000	6.000000

```
tips.info()
```

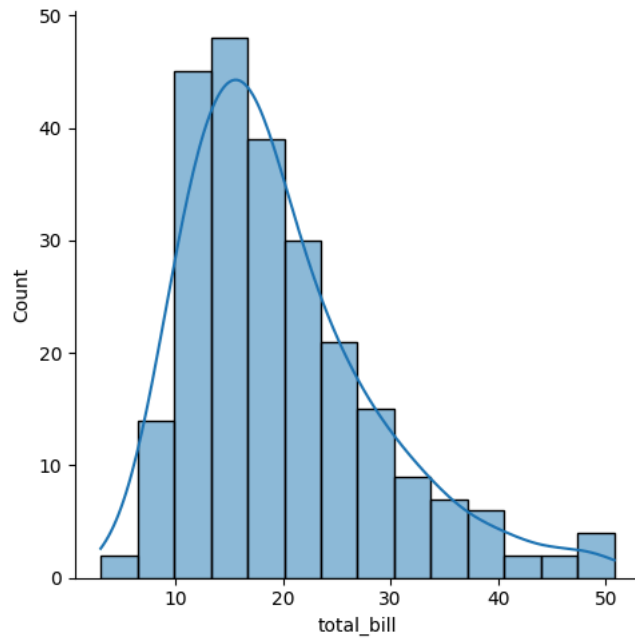
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 244 entries, 0 to 243
Data columns (total 7 columns):
#   Column      Non-Null Count  Dtype
---  -
0   total_bill  244 non-null   float64
1   tip         244 non-null   float64
2   sex         244 non-null   category
3   smoker      244 non-null   category
4   day         244 non-null   category
5   time        244 non-null   category
6   size        244 non-null   int64
dtypes: category(4), float64(2), int64(1)
memory usage: 7.4 KB
```

```
tips.isnull().sum()
```

	0
total_bill	0
tip	0
sex	0
smoker	0
day	0
time	0
size	0
dtype:	int64

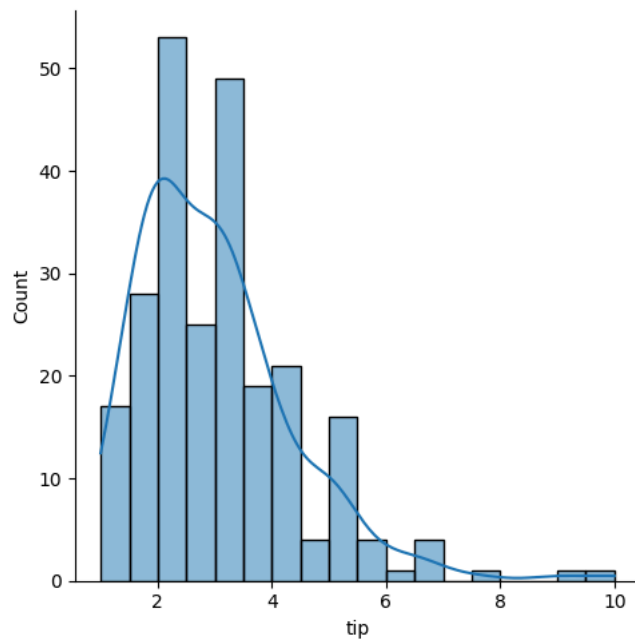
```
sns.displot(tips.total_bill,kde=True)
```

```
<seaborn.axisgrid.FacetGrid at 0x7d73c87865d0>
```



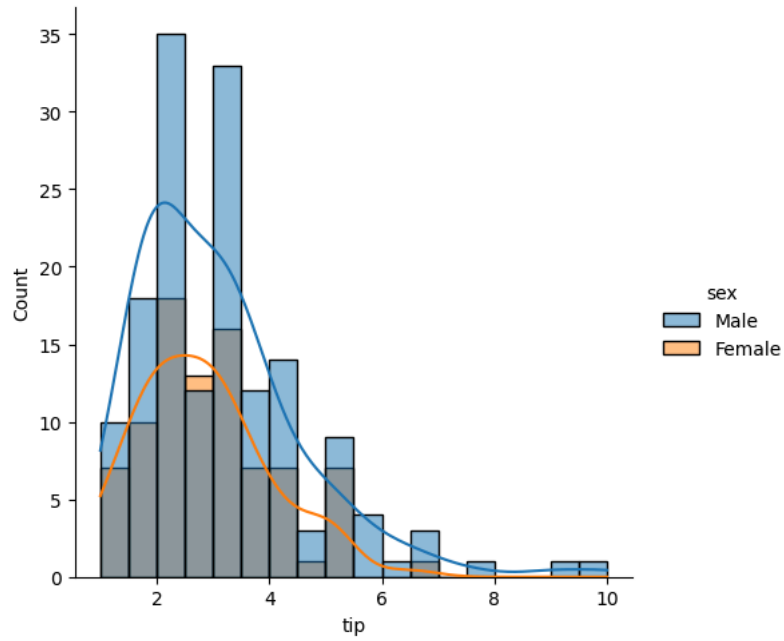
```
sns.displot(tips.tip,kde=True)
```

```
<seaborn.axisgrid.FacetGrid at 0x7d73c87e65d0>
```



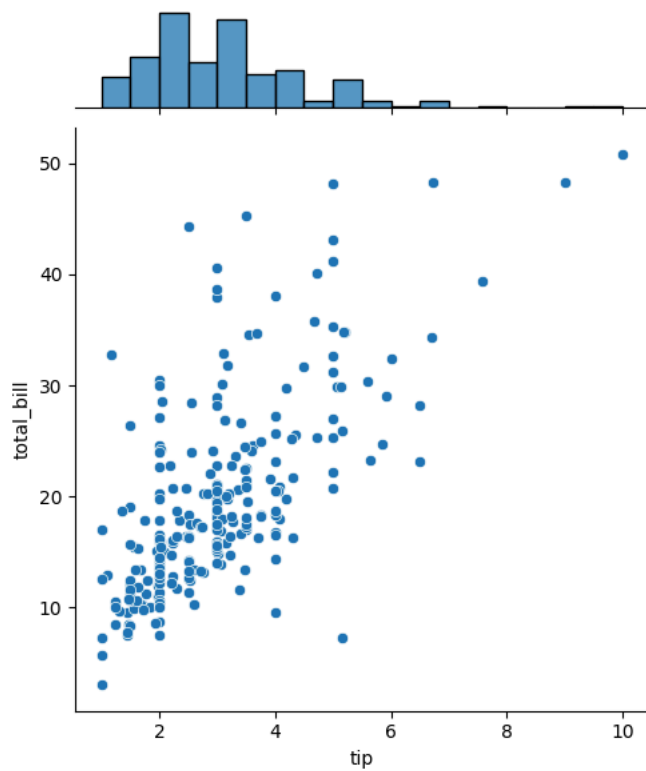
```
sns.displot(data=tips, x='tip', hue='sex', kde=True)
```

```
<seaborn.axisgrid.FacetGrid at 0x7d73c649ac60>
```



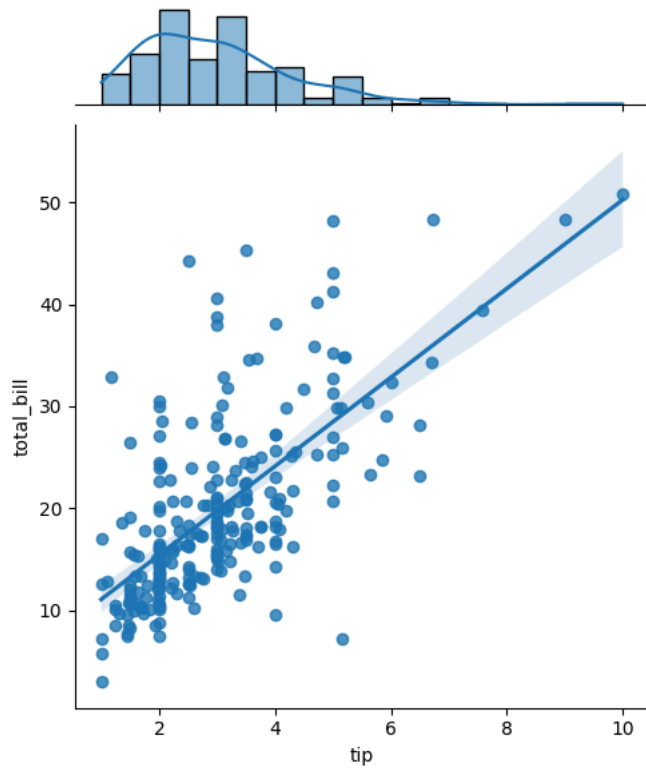
```
sns.jointplot(x=tips.tip,y=tips.total_bill)
```

```
<seaborn.axisgrid.JointGrid at 0x7d73c407b500>
```



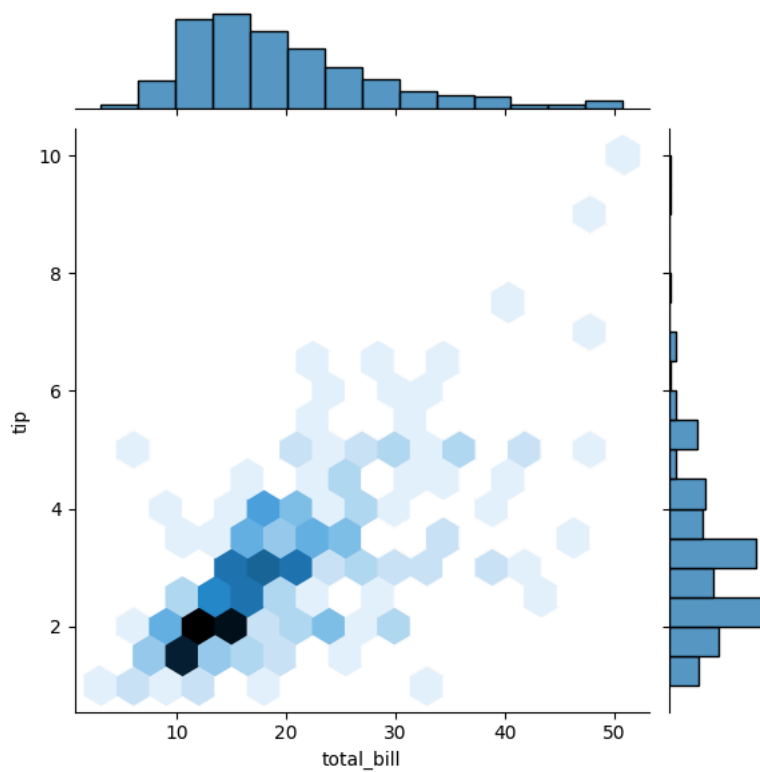
```
sns.jointplot(x=tips.tip,y=tips.total_bill,kind="reg")
```

```
<seaborn.axisgrid.JointGrid at 0x7d73c3ea94c0>
```



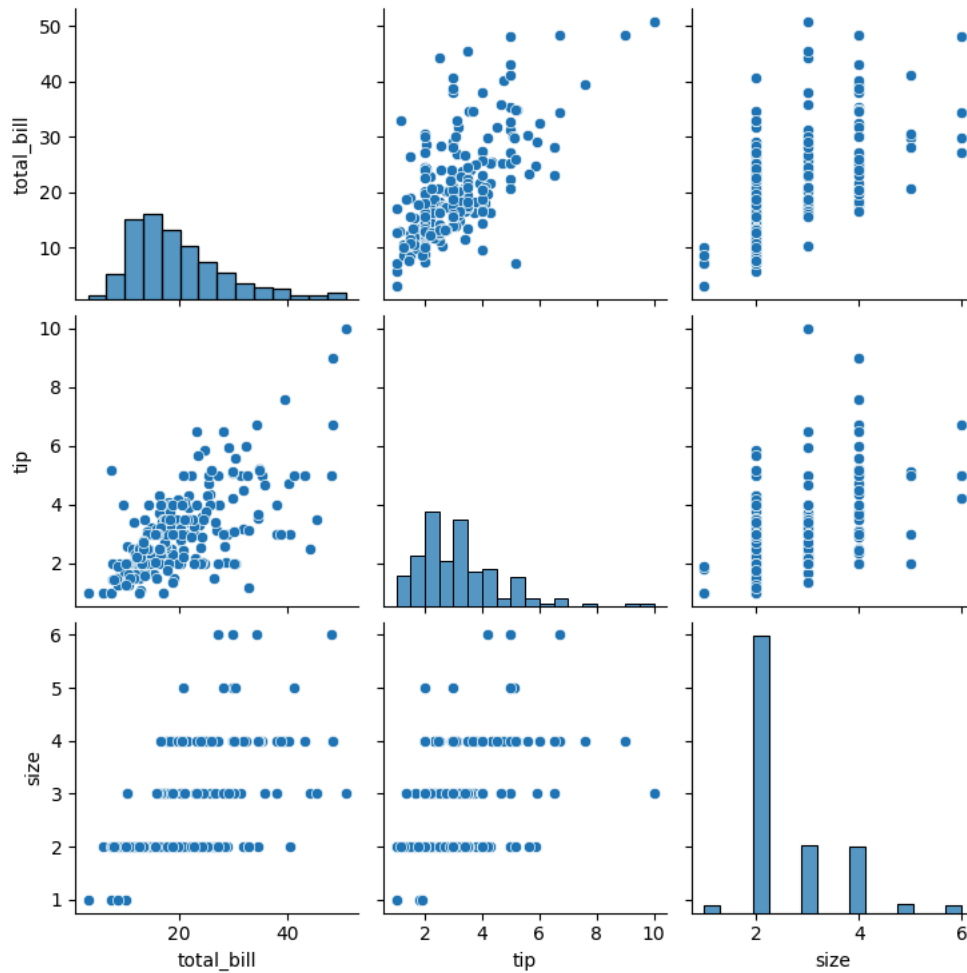
```
sns.jointplot(x=tips.total_bill,y=tips.tip,kind='hex')
```

```
<seaborn.axisgrid.JointGrid at 0x7d73c24c1e80>
```



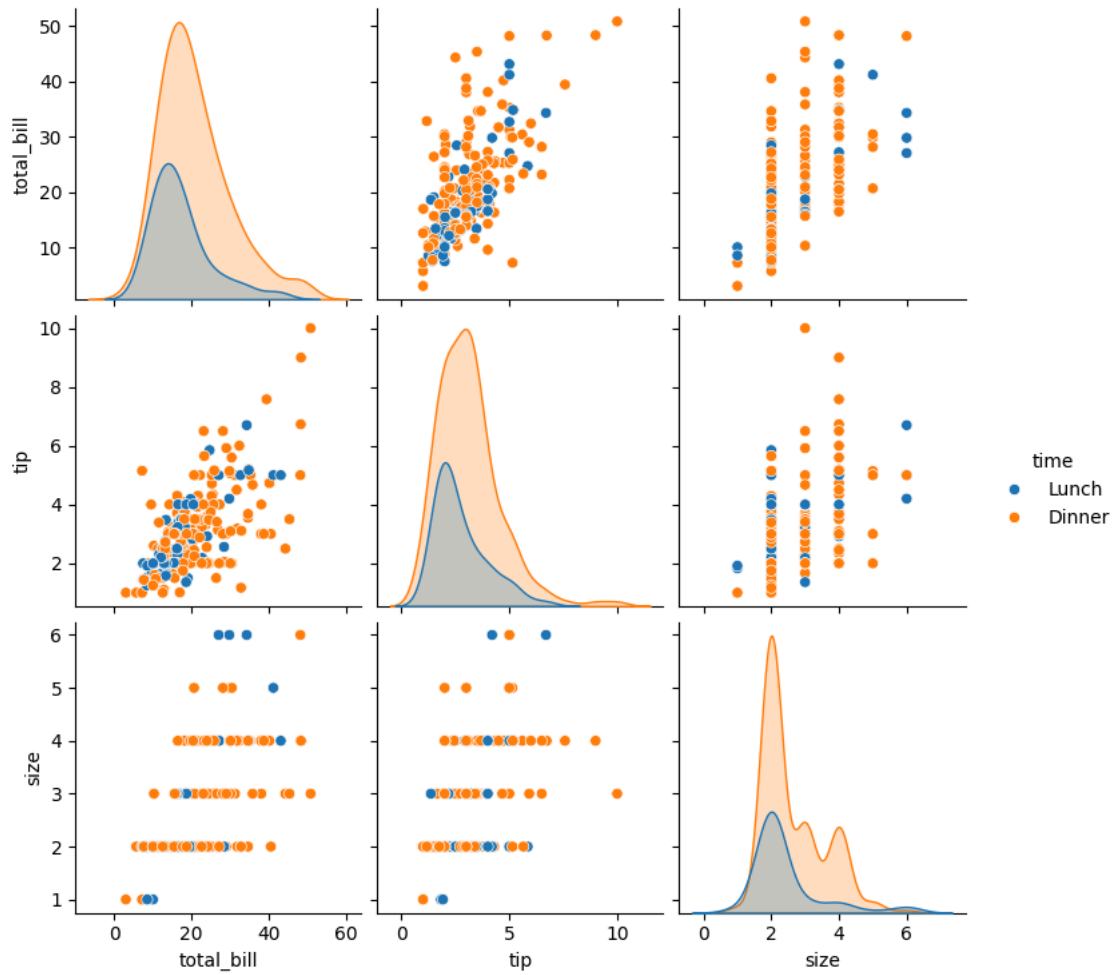
```
sns.pairplot(tips)
```

```
<seaborn.axisgrid.PairGrid at 0x7d73c242a1b0>
```



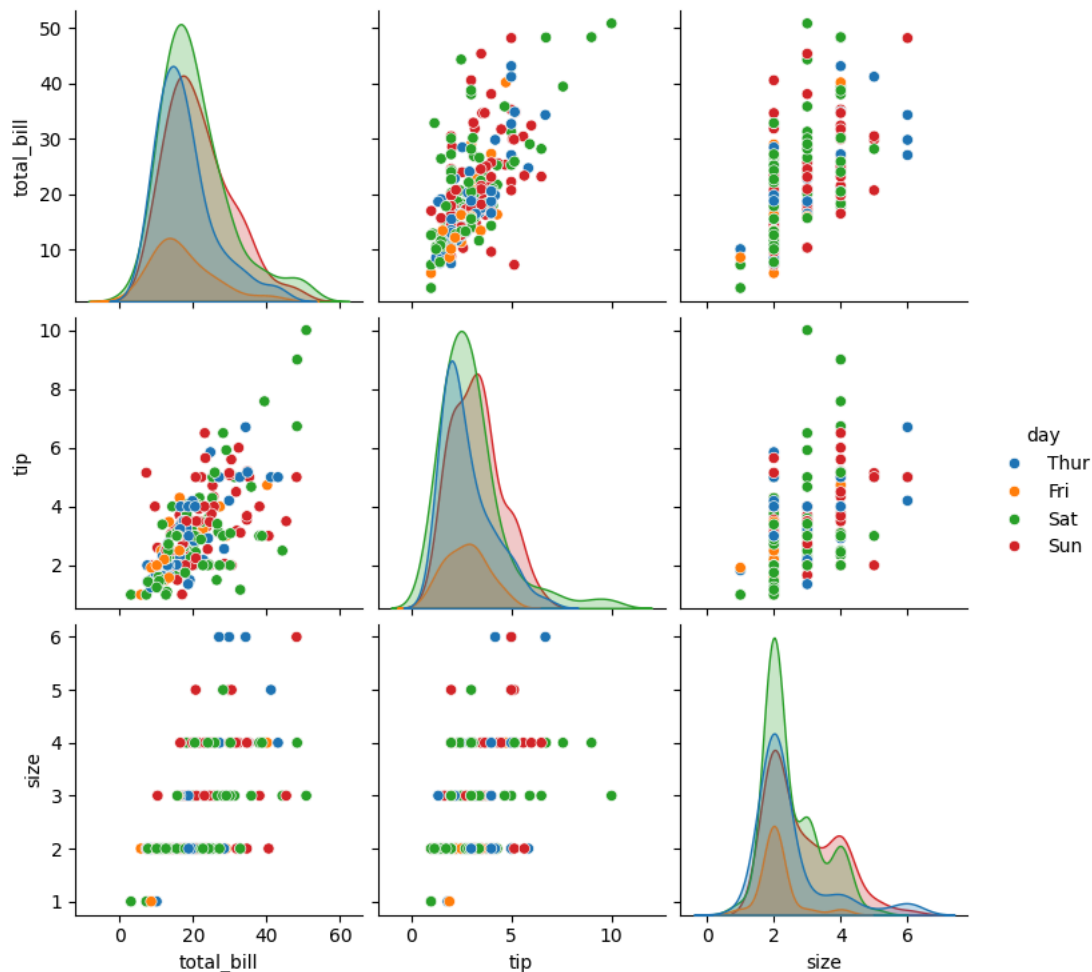
```
sns.pairplot(tips, hue='time')
```

<seaborn.axisgrid.PairGrid at 0x7d73c21434a0>



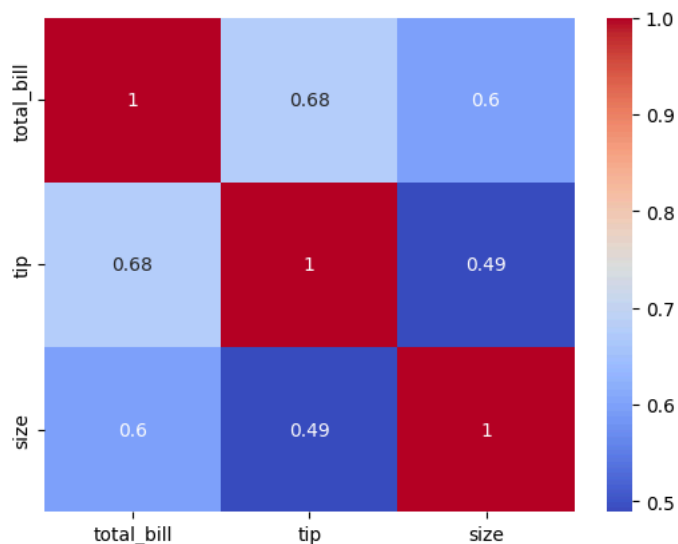
```
sns.pairplot(tips, hue='day')
```

```
<seaborn.axisgrid.PairGrid at 0x7d73c204bf50>
```



```
sns.heatmap(tips.corr(numeric_only=True),annot=True,cmap='coolwarm')
```

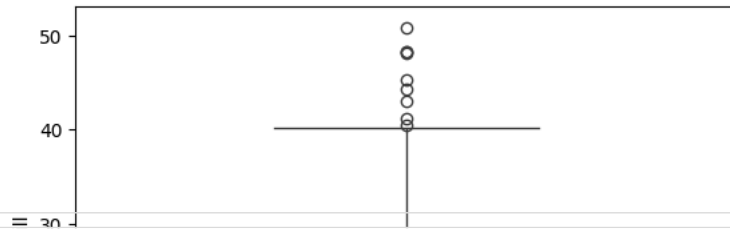
```
<Axes: >
```



```
sns.boxplot(tips.total_bill)
```

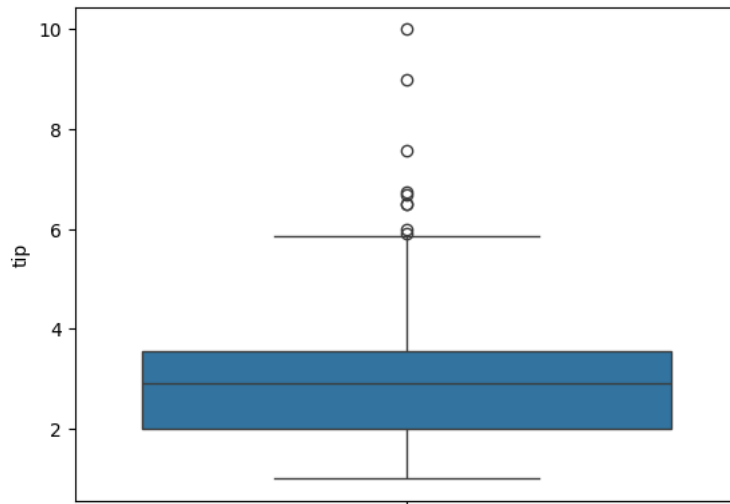


```
<Axes: ylabel='total_bill'>
```



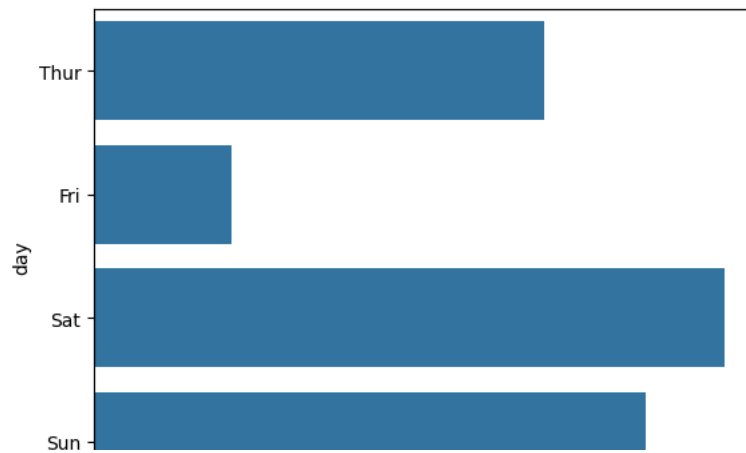
```
sns.boxplot(tips.tip)
```

```
<Axes: ylabel='tip'>
```



```
sns.countplot(tips.day)
```

```
<Axes: xlabel='count', ylabel='day'>
```



Result:

Hence, a Python program to perform Exploratory Data Analysis has been executed successfully, providing valuable insights into the dataset's structure, relationships, and patterns for further processing.